Azure HPC Batch Computing

Empowering High-Performance Workloads

Introduction to HPC Batch

- **What is Azure HPC Batch?**
  - Managed service for large-scale parallel and batch workloads.
  - Automates scheduling, resource scaling, and job execution.
- **Use Cases**
  - Engineering simulations
  - Financial risk modeling
  - Weather forecasting
  - Image analysis/rendering

Benefits of Azure Batch

Scalability: Automatically scales resources based on workload demand.

Ease of Use: Simplified job scheduling and workload management.

Integration: Works with Azure services like Blob Storage, Virtual Machines, and more.

Flexibility: Supports Linux and Windows workloads, and containerized applications.

Cost Management: Leverage low-priority VMs and Spot VMs for cost optimization.

Getting Started

Prerequisites

Azure subscription.

Familiarity with HPC workloads and tools.

Resources

Azure Portal, SDKs (Python, .NET), and REST APIs.

Quickstart

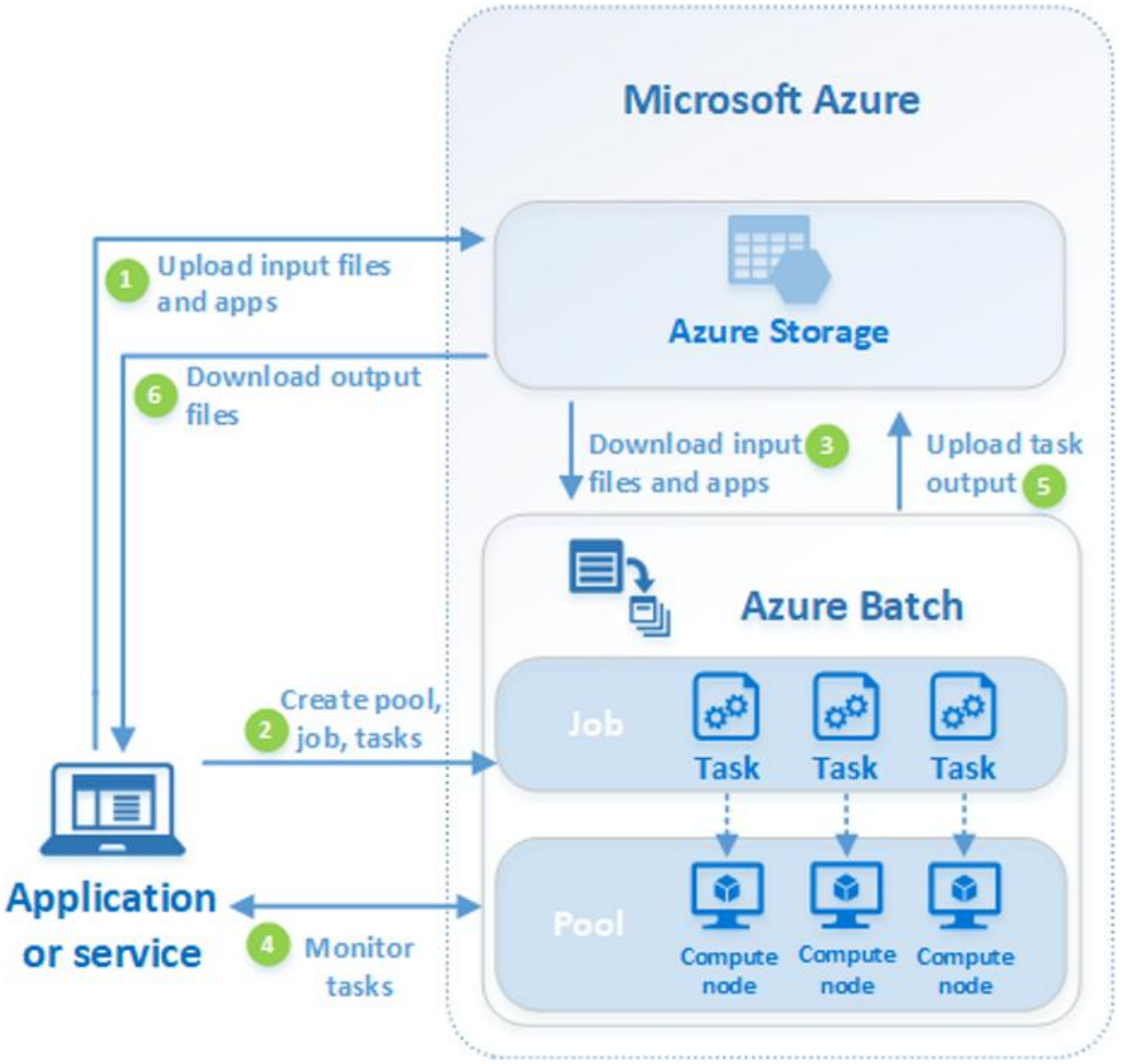
Create a Batch account.

Configure a pool of compute nodes.

Submit your first job.

How it works

1. Upload Files
2. Create Batch Pool
3. Download Files
4. Monitor Task Execution
5. Upload Task Output
6. Download Files



https://learn.microsoft.com/en-us/azure/batch/batch-technical-overview

Components of Azure Batch

Pool

- Set of compute nodes (VMs)

Job

- Group of one or more tasks

Task

- Tasks execute a specific operation or action

Storage

- Storing of resource files and output files

Integration with Azure Ecosystem

Compute Options: Support for GPUs, FPGAs, and high-memory VMs.

Storage Integration: Azure Blob Storage, Azure Data Lake.

Networking: Private link, VNet integration for secure operations.

Monitoring & Diagnostics: Azure Monitor, Log Analytics.

Best Practices

01

Optimize for costs with Spot VMs.

02

Leverage parallel processing for efficiency.

03

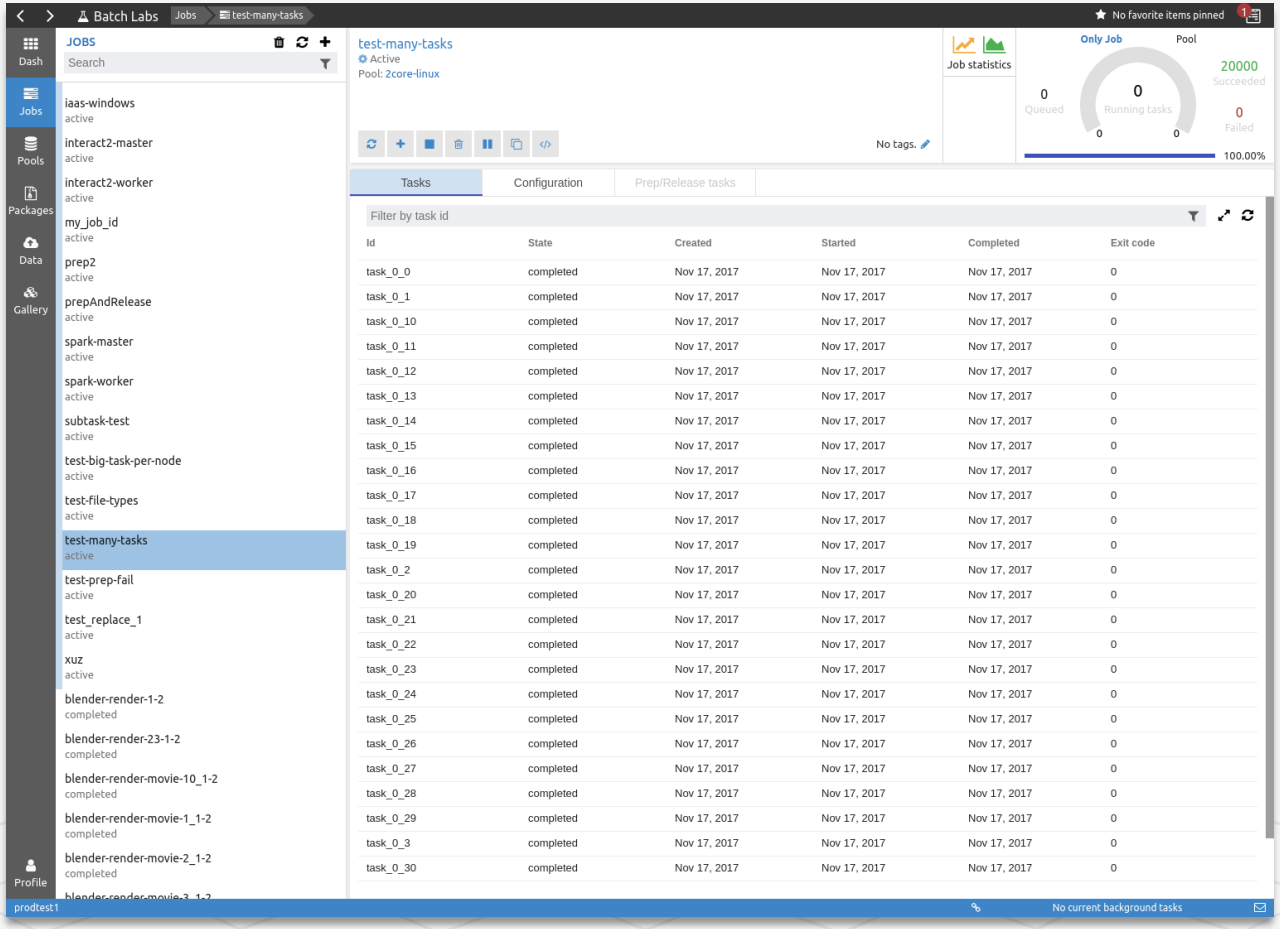
Use auto-scaling to adapt to workload needs.

04

Secure data with encryption and network isolation.

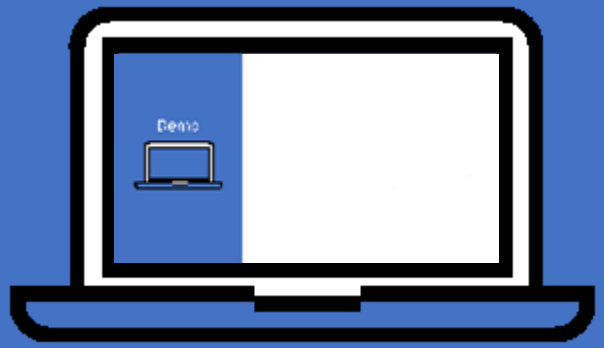
Batch Explorer

Tool to manage your Azure Batch accounts



https://azure.github.io/BatchExplorer/

Demo



Azure Bach

Summary

- Why Azure HPC Batch?

  Scalable, cost-efficient, and easy to use.
  No need to be a pro at Slurm the **Batch Management API** for managing Pools or Jobs.

Alternatives

- Azure CycleCloud
  - Tool for orchestrating and managing High Performance Computing (HPC) environments on Azure

Resources

Azure Batch Documentation

https://learn.microsoft.com/en-us/azure/batch/

Run a parallel workload with Azure Batch using the Python API

https://learn.microsoft.com/en-us/azure/batch/tutorial-parallel-python

Link to Slides in GitHub

https://github.com/danielecolon/Azure-Batch